Chapter the Third

Elements of Mechanics

illustrated by figures of the simplest engines

Of the Cart

A load placed upon wheels is moved by a force much less than its weight, because the wheels turn instead of rubbing along the ground. The force F has then only to overcome the small friction at the axles, together with the resistance of the ground to the rolling of the rims.

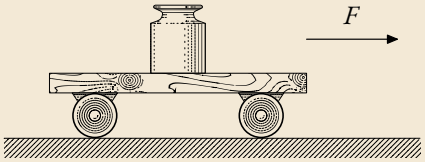


Fig. 1. A loaded cart drawn along level ground.

Of the Lever

A bar resting upon a prop, and carrying weights at its two ends, is at rest when the weights are to one another inversely as their distances from the prop. Thus in Fig. 2 the greater weight, being nearer the prop, balances the smaller placed farther from it; and the rule is

$$W_1\, a = W_2\, b.$$

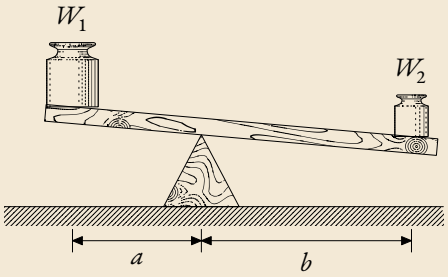


Fig. 2. Two weights in balance upon a lever.

Of the Pendulum

A heavy ball hung by a thread, and drawn aside through the angle θ, returns toward the lowest point and passes beyond it, so that the motion is repeated. For small excursions the time of one swing depends upon the length l alone, and not upon the weight of the ball nor upon the width of the arc; whence the pendulum serves to divide the day into equal parts.

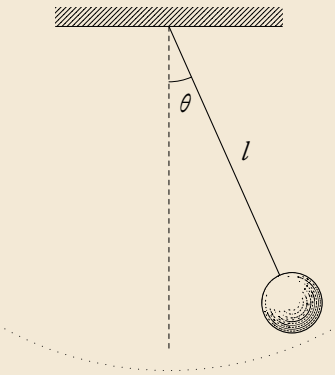


Fig. 3. A pendulum drawn aside from the vertical.

Of Weight and Measure

The figures are drawn by the same MetaPost library that made the plates of the preceding chapters, but here they are produced inside the page itself: LuaTeX's built-in MetaPost interpreter renders each one at the place where it stands in the column, and the letters upon them are set in the type of the text. The wood grain, the hatching of the ground, the shading of the ball, and the standing weights are stock parts of the library; the arrows, dimension lines, and dotted arc are plain MetaPost added around them.

A page of this kind was once the work of an engraver and a compositor in turn. Here the two are done together, and a change of a single length in the source moves the weights, the arrows, and the letters at once.

To be continued in the next number